

Tribhuvan University
Faculty of Humanities and Social Sciences
Patan Multiple Campus
Masters of Computer Application (MCA)
First Semester
Advanced Database Management System (MCA-503)

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Submitted By:

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Department - Department of MCA

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Submitted Date: 2083/04/16

Laboratory Work Table of Content

S.No.	Laboratory Work	Page No.	Remarks
1.	Installation of oracle 21c DBMS software, Create Database schema for given application and insert the data into the database as required for the laboratory work.		Individual
2.	Execute SQL data definition Language (DDL) commands (CREATE, ALTER, DROP, TRUNCATE, RENAME) to manage Database in the database.		Individual
3.	Execute SQL data manipulation Language (DML) Commands (SELECT, INSERT, UPDATE, DELETE) to manipulate data in the database.		Individual
4.	Execute SQL data control Language (DCL) commands (GRANT, REVOKE) to control the access of data in the database.		Individual
5.	Execute SQL transaction Control Language (TCL) commands (COMMIT, ROLLBACK, SAVEPOINT) to control transactions of data in the database		Individual
6.	Write Queries using Arithmetic, relational and Logical operators to manipulate the data in the given database.		Individual
7.	Illustrate enforcing integrity constraints (Domain, Key constraints (Primary/Foreign keys), not null, unique, default, Check) in the given database data.		Individual
8.	Write queries to use Set operators (ANY, ALL, IN, INTERSECT, UNION) to perform different operations in the given database data.		Individual
9.	Write queries using Date and Time functions, string functions and Arithmetic functions.		Individual
10.	Implement queries using Aggregate functions, Ordering and Grouping (COUNT, SUM GROUPBY and HAVING.)		Individual
11.	Write SQL queries to implement the various types of joins such as Inner and Outer Join		Individual
12.	Implement database views by creating, modifying, querying, and dropping views for efficient data access.		Individual

13.	Implement SQL subqueries, including single-row, multiple-row, multiple-column, and correlated subqueries for data retrieval and manipulation.		Individual
14.	Write a PL/SQL program to illustrate the Conditional Statements in PL/SQL.		Individual
15.	Write a PL/SQL program to illustrate the Iterative Statements in PL/SQL.		Individual
16.	Write a PL/SQL program to illustrate the Sequential Control in PL/SQL.		Individual
17.	Write a PL/SQL program to demonstrate the use of an Implicit Cursor.		Individual
18.	Write a PL/SQL program to demonstrate the use of an Implicit Cursor.		Individual
19.	Write a PL/SQL program to illustrate Exception Handling using Predefined Exceptions.		Individual
20.	Write a PL/SQL program to illustrate Exception Handling using User-Defined Exceptions.		Individual
21.	Create PL/SQL stored procedures using IN, OUT, and IN OUT parameters to illustrate modularity.		Individual
22.	Write a PL/SQL procedures to illustrate the CRUD operations with the transaction control like commit, rollback in the database using stored procedure.		Individual
23.	Create user-defined PL/SQL functions of your choice for the given database and demonstrate their use in SQL queries and PL/SQL blocks.		Individual
24.	Write a PL/SQL program to create INSERT, UPDATE, and DELETE triggers for the given database.		Individual
25.	Implement indexes and sequences for the given database and demonstrate their use in improving query performance and generating unique key values.		Individual
26.	Object and Object-Relational Databases (Oracle DB): - Create object types using Oracle Object Types. - Create object tables based on user-defined object types.		Group of at most 2 members

	- Insert, update and delete objects from object tables. - Create nested object attributes. - Create collection data types (VARRAY and Nested Table). - Implement object references (REF). - Create object methods (Member Functions and Procedures). - Query object tables using dot notation. - Implement inheritance using UNDER clause. - Write Object SQL queries for object retrieval.		
27.	Query Processing (Oracle DB): - Translate SQL queries into Relational Algebra. - Compare execution plans using EXPLAIN PLAN. - Perform different JOIN operations. - Implement aggregation queries. - Compare GROUP BY with HAVING. - Compare Nested Loop Join and Hash Join. - Analyze ORDER BY using large datasets. - Simulate pipelined query execution. - Execute queries with indexes and without indexes. - Compare query execution cost using AUTOTRACE.		Group of at most 2 members
28.	Query Optimization (Oracle DB): - Draw query trees for SQL statements. - Convert SQL into relational algebra trees. - Optimize queries using heuristic rules. - Compare optimized and non-optimized queries. - Analyze query execution plans. - Create indexes and observe optimization. - Measure execution time before and after indexing. - Compare different join algorithms. - Perform cost-based optimization using Oracle Optimizer. - Optimize complex nested queries.		Group of at most 2 members

29.	Distributed Database Concepts (Oracle DB): - Install two database instances. - Create Database Links. - Execute distributed SQL queries. - Perform horizontal fragmentation. - Perform vertical fragmentation. - Implement data replication. - Execute distributed joins. - Simulate distributed transactions. - Demonstrate two-phase commit.		Group of at most 3 members
30.	NOSQL (MongoDB): - Install MongoDB. - Create databases and collections. - Insert JSON documents. - Query documents. - Update documents. - Delete documents. - Perform aggregation pipeline operations. - Create indexes. - Perform embedded document operations.		Individual
31.	Big Data (Hadoop): - Downloading and Installing Hadoop: Understanding different Hadoop modes, Startup Scripts, Configuration files - Hadoop Implementation of file management tasks, such as Adding files and directories Retrieving files and Deleting files. - Implement of Matrix Multiplications with Hadoop Map Reduce - Run a basic Word Count Map Reduce program to understand Map Reduce Paradigm. - YARN Job Monitoring and Cluster Management illustration - Installation of HIVE along with practice examples		Group of at most 3 members

	- Installation of HBase, Installing thrift along with Practice examples. - Practice importing and exporting data from various databases.		
32.	Enhanced Data Models (Oracle DB): - Implement active database concepts by creating INSERT, UPDATE, and DELETE triggers to automate database operations and enforce business rules. - Implement temporal database concepts by designing temporal tables and executing queries to store and retrieve historical data. - Implement spatial database concepts by configuring Oracle Spatial/PostGIS, creating spatial objects, and executing spatial queries. - Implement multimedia database concepts by storing, managing, and retrieving multimedia objects in the database. - Implement deductive database concepts by writing recursive SQL queries and developing rule-based database applications.		optional
33.	Project: Design and develop your own database for a real-world application of your choice. The database must contain at least five related tables and include database objects such as views, stored procedures, user-defined functions, etc. Develop an application that uses the database to demonstrate CRUD operations. Also make sure application uses each database object you have created.		Project can be completed in a group of 4 members at most.

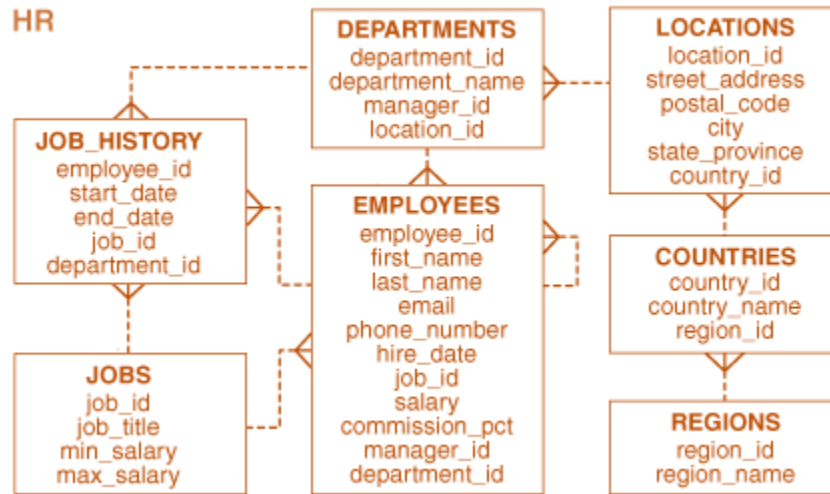
Tools Used: Oracle 21c, SQL Server, Postgres SQL, NOSQL- MongoDB, BIG data – Hadoop, HDFS, MapReduce, Hive, and HBase

Experiment 1 - DML

1.1. Objective: To understand use of SQL data manipulation Language (DML) commands (SELECT, INSERT, UPDATE, DELETE) for manipulation of data in a Database.

1.2. Tools Used: Oracle 21c, SQL Developer or VSCode (with oracle SQL developer for Vscode Extension)

1.3. Database / Table Used:



1.4. Theory: Data Manipulation Language (DML) is a subset of SQL used to manipulate the data stored in database tables. DML commands allow users to insert, retrieve, modify, and delete records without changing the table structure.

Command	Description
INSERT	Adds one or more records into a table.
SELECT	Retrieves data from one or more tables.
UPDATE	Modifies existing records in a table.
DELETE	Removes one or more records from a table.

1.5. Query to Database

1.5.1. INSERT

➤ **Syntax:**

```
INSERT INTO table_name (column1, column2, ...)
VALUES (value1, value2, ...);
```

➤ **Query:**

```
INSERT INTO employees (employee_id, first_name, last_name, email,
phone_number, hire_date, job_id, salary, department_id)
```

```
VALUES(999, 'Tek', 'Yogi', 'tek@gmail.com', '9848751271', SYSDATE,
'IT_PROG', 6000, 60);
```

➤ Database before query:

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	1.515.555.8198	17-JUN-13	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Yang	NYANG	1.515.555.8221	21-SEP-15	AD_VP	17000	(null)	100	90
3	102	Lex	DeHaan	LDEHAAN	1.515.555.8144	13-JAN-11	AD_VP	17000	(null)	100	90
4	103	Alexander	Russell	ARUSSEL	1.590.555.8293	03-OCT-16	IT_PROG	9000	(null)	102	60
5	104	Bruce	Smith	BSMITH	1.590.555.8204	21-MAY-17	IT_PROG	6000	(null)	103	60
6	105	David	Williams	DWILLIAMS	1.590.555.8305	25-JUN-15	IT_PROG	4000	(null)	102	60
7	106	Valli	Pataballa	VPATABALL	1.590.555.8306	05-FEB-16	IT_PROG	4000	(null)	103	60
8	107	Doreen	Greenberg	DGREENB	1.590.555.8307	07-FEB-17	IT_PROG	4200	(null)	103	60
9	108	Nancy	Greenberg	NGREENB	1.515.555.8308	17-AUG-12	FI_MGR	12000	(null)	101	100
10	109	Daniel	Favariet	DFAVIET	1.515.555.8309	18-AUG-12	FI_ACCOUNT	9000	(null)	108	100
11	110	John	Chen	JCHEN	1.515.555.8310	28-SEP-15	FI_ACCOUNT	8200	(null)	108	100
12	111	James	Sciarra	JSCIARRA	1.515.555.8311	30-SEP-16	FI_ACCOUNT	7700	(null)	108	100
13	112	Jose Manuel	Urman	JMURMAN	1.515.555.8312	07-MAR-16	FI_ACCOUNT	7900	(null)	108	100
14	113	Luis	Popp	LPOPP	1.515.555.8313	07-DEC-17	FI_ACCOUNT	6900	(null)	108	100
15	114	Den	Li	DLI	1.515.555.8314	07-DEC-12	PU_MGR	11000	(null)	109	30
16	115	Alexander	Khoo	AKHOO	1.515.555.8315	18-MAY-13	PU_CLERK	3100	(null)	114	30
17	116	Shelli	Baida	SBIDA	1.515.555.8316	24-DEC-15	PU_CLERK	2900	(null)	114	30

➤ Database After query:

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	99	Tek	Yogi	tek@gmail.com	9848751271	21-SEP-15	IT_PROG	6000	(null)	(null)	90
2	100	Steven	King	SKING	1.515.555.8198	17-JUN-13	AD_PRES	24000	(null)	(null)	90
3	101	Neena	Yang	NYANG	1.515.555.8221	21-SEP-15	AD_VP	17000	(null)	100	90
4	102	Lex	DeHaan	LDEHAAN	1.515.555.8144	13-JAN-11	AD_VP	17000	(null)	100	90
5	103	Alexander	Russell	ARUSSEL	1.590.555.8293	03-OCT-16	IT_PROG	9000	(null)	102	60
6	104	Bruce	Smith	BSMITH	1.590.555.8204	21-MAY-17	IT_PROG	6000	(null)	103	60
7	105	David	Williams	DWILLIAMS	1.590.555.8305	25-JUN-15	IT_PROG	4000	(null)	102	60
8	106	Valli	Pataballa	VPATABALL	1.590.555.8306	05-FEB-16	IT_PROG	4000	(null)	103	60
9	107	Doreen	Greenberg	DGREENB	1.590.555.8307	07-FEB-17	IT_PROG	4200	(null)	103	60
10	108	Nancy	Greenberg	NGREENB	1.515.555.8308	17-AUG-12	FI_MGR	12000	(null)	101	100
11	109	Daniel	Favariet	DFAVIET	1.515.555.8309	18-AUG-12	FI_ACCOUNT	9000	(null)	108	100
12	110	John	Chen	JCHEN	1.515.555.8310	28-SEP-15	FI_ACCOUNT	8200	(null)	108	100
13	111	James	Sciarra	JSCIARRA	1.515.555.8311	30-SEP-16	FI_ACCOUNT	7700	(null)	108	100
14	112	Jose Manuel	Urman	JMURMAN	1.515.555.8312	07-MAR-16	FI_ACCOUNT	7900	(null)	108	100
15	113	Luis	Popp	LPOPP	1.515.555.8313	07-DEC-17	FI_ACCOUNT	6900	(null)	108	100
16	114	Den	Li	DLI	1.515.555.8314	07-DEC-12	PU_MGR	11000	(null)	109	30
17	115	Alexander	Khoo	AKHOO	1.515.555.8315	18-MAY-13	PU_CLERK	3100	(null)	114	30
18	116	Shelli	Baida	SBIDA	1.515.555.8316	24-DEC-15	PU_CLERK	2900	(null)	114	30
19	117	Sigal	Thirumangalakudi	STHIRUM	1.515.555.8317	24-SEP-15	PU_CLERK	3000	(null)	114	30
20	118	Guy	Himuro	GHHIMURO	1.515.555.8318	25-11-2013	PU_CLERK	2000	(null)	114	30
21	999	Tek	Yogi	tek@gmail.com	9848751271	21-SEP-15	IT_PROG	6000	(null)	(null)	90

[Note: show the effect of query execution on the database]

1.5.2. UPDATE

[Query and output for the update command is placed here]

1.5.3. DELETE

[Query and output for the delete command is placed here]

1.5.4. SELECT

[Query and output for the select command is placed here]

1.6. Discussion:

This lab demonstrated the full data-manipulation lifecycle on a single table. INSERT populated the table with both complete rows and a partial row (illustrating that unspecified columns default to NULL). SELECT is used in increasing complexity retrieving all data, filtering with WHERE, sorting with ORDER BY, and summarizing with GROUP BY and aggregate functions (COUNT, AVG) showing how the same command scales from simple retrieval to analytical queries. UPDATE showed that the WHERE clause is what determines the scope of the change: a condition matching a unique key affects one row, while a condition matching a shared attribute value (city = 'Kathmandu') affects multiple rows at once. DELETE behaved analogously, with the important distinction that DELETE removes rows but preserves the table definition, unlike a DDL command such as DROP TABLE or TRUNCATE TABLE. Because all four commands operate within a transaction, COMMIT was used after each set of changes to make them permanent; without it, a ROLLBACK could have reverted the changes.

1.7. Conclusion:

This lab provided practical experience executing all four core DML commands SELECT, INSERT, UPDATE, and DELETE reinforcing how each command manipulates table data, how the WHERE clause governs the scope of UPDATE/DELETE, and how SELECT can range from simple row retrieval to filtered, sorted, and aggregated queries.